

WFH07BluetoothCommunicationProtocol

Version<A4.0>

Versionhistory

Version	Preparedby	Dateof Preparation	Reviewedby	Date	Status
A1.0	Yuhui	2024.03.27	Tony	2024.03.29	
A3.0	Yuhui	2024.11.04	Tony	2024.11.11	Add flashing and remote control
A4.0	Yuhui	2024.12.24	Tony	2024.12.24	Modify flashing and remote control

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1. Electronic Hardware Diagram



Figure 1

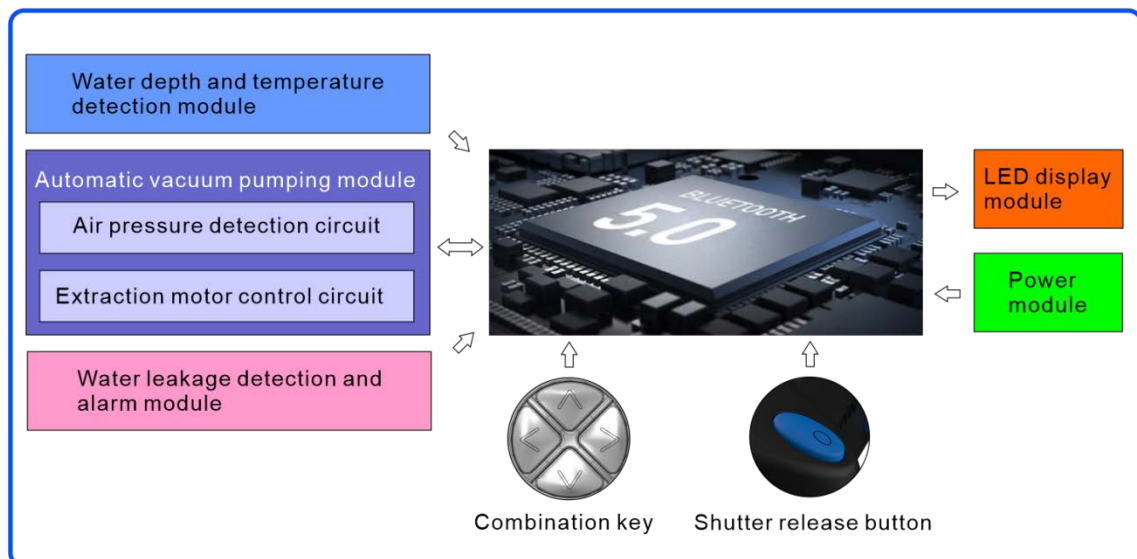


Figure2

Module	Function	Remarks
BluetoothLowEnergy	Realizedatacommunicationandexchange between smarthousingandmobileAPP accordingtoBluetoothprotocol.	Version: BLE5.0
Powermodule	Includingthepowersupplyvoltage regulator controlmodule, lithiumbatterychargeand discharge managementmodule, andlithium batteryvoltage detectionmodule. Managethe chargeanddischarge, powerstabilityand voltage detection oflithiumbattery.	Pleasereferto Section 5. 1 for details.
Automaticvacuum pumpingmodule	IncludingExtractionmotorcontrolcircuitandAir pressuredetectioncircuit. Bycontrollingthetartand stopofthemotor, theairinsidethewaterproofcasingis extractedtoformanegativeairpressurestate, soasto simulatethesealingconditionofthecasing. Duringthe pumpingprocess, thecurrentairpressurechangeinside thewaterproofcaseiscontinuouslydetected, anddata analysisisusedtodeterminewhetherthereisariskof waterleakage.	Topredictwater leakageby calculatingair pressure, referto Section 5.4for details.
Keymodule	Includingthecombinationbuttonandshutter button, thecurrentlypressedbuttoninformation isdetected, processedbytheBluetoothmodule insidetheproduct, andthen theinformationis senttothemobilephonethroughBluetooth wireless transmissionintheformofaspecific code	TheBluetooth servicecanobtain dataforthekey, pleasereferto Section 5.3for details.
Waterdepthand temperaturedetection module	Thewaterpressureandwatertemperatureof thecurrentenvironmentaredetectedbythe waterpressuresensorinstalledinthecasing, andthedataiswirelesslytransmittedtothe mobilephonethroughBluetooth.	TheBluetooth servicecanobtain dataforthesensor, seesection 5.4for details.
LEDdisplaymodule	ThecurrentBluetoothstatus, chargingstatus andsealingstatusofthecasingcanbeindicated bythedifferentstatesdisplayedthroughthe LEDindicator.	Redlighton: charging Greenlighton: charging complete Bluelighton: Bluetooth isconnected Bluelightflashing: Bluetoothiswaitingto connect.

Table 1

Remarks:

(1) The data sent from the casing to the mobile phone via Bluetooth wireless transmission:

- A. Battery level data
- B. Key detection data
- C. Data of water pressure and temperature detected by water pressure sensor
Data from the barometric sensor
- D. Data of device parameters (e.g., manufacturer, firmware version, etc.)
- E. Air extraction cover signal

(2) Data sent from mobile phone to the casing through Bluetooth wireless transmission:

- A、Data of the air extraction motor control command issued from the mobile APP
- B、Data of the solenoid valve control command issued from the mobile APP
- C、Data of the Trigger flash control command issued from the mobile APP
- D、OTA firmware upgrade data

(3) How to use this product for automatic vacuum pumping

When the user needs to turn on vacuum detection, the following steps need to be taken:

1. Confirm opening the suction cover for motor exhaust
2. Open the solenoid valve to prevent the exhausted gas from entering the interior of the waterproof shell again
3. Turn on the motor and start pumping air
4. When the specified air pressure (Can use the internal air pressure value of the phone, Hardware can cancel the pressure sensor) is reached, stop pumping and turn off the motor
5. Confirm that the suction cover is closed, and the detection of small waterproof leaks is not accurate
6. Close the solenoid valve so that the waterproof shell can be easily opened when opening the suction cover. Close the solenoid valve in advance to save power
7. Check for more than 5 minutes and determine if there is any air or water leakage based on the air pressure

2. Electronic hardware specifications

Bluetooth	Protocol	Transmitting power	Communication distance	Reception sensitivity	Voltage
	BLE5.0	4dBm	100m	-95dBm	1.8~3.6V
Battery	Battery type	Capacity	Voltage	Charge and discharge times	Charging time
	Lithium polymer	300mah	3.7V	500 times	Approx. 2H
Charging type	TYPE C 5V/ 1A				
Run time	> 24H (estimated)				
Standby time after shutdown	> 60D (estimated)				
Electric pumping times	100 times (estimated)				
Water depth detection	0~ 1400Kpa Resolution 50 Kpa				
Air pressure detection	30 Kpa ~120 Kpa Resolution 5Kpa				
Times of button use	> 10000 times				

Table2

3. Bluetooth working diagram

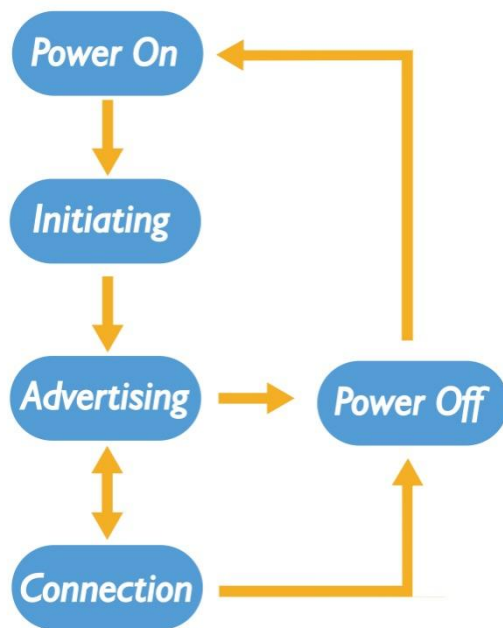


Figure2

4. Bluetooth technical parameters

4. 1 Bluetooth Advertising Parameters

Advertising parameters	Contents	Remarks
Advertising_ Type	ConnectableUndirected EventType	
Advertising_ Name	DIVE IT	
Advertising_ Data	MAC address, 6 bytes	
Advertising_ Address_ Type	StaticDeviceAddress	
Advertising interval	200MS	
Advertising duration	180S	

Table 3

4.2 Bluetoothconnectionparameters

Connectionparameters	Contents	Remarks
minconninterval	Canbespecified	Current:25ms
maxconninterval	Canbespecified	Current:125ms
SlaveLatency	Canbespecified	Current: 0
SupervisionTimeout	Canbespecified	Current: 4S

Table4

To modify these parameters, the iPhone must meet the following conditions:

- ① $\text{Max_conn_interval} * (\text{SlaveLatency} + 1) \leq 2\text{s}$
- ② $\text{Min_conn_interval} \geq 20\text{ms}$
- ③ $\text{Min_conn_interval} + 20\text{ ms} \leq \text{max_conn_interval}$
- ④ $\text{SlaveLatency} \geq 4$
- ⑤ $\text{SupervisionTimeout} \leq 6\text{s}$
- ⑥ $\text{max_conn_interval} * (\text{SlaveLatency} + 1) * 3 < \text{SupervisionTimeout}$

5. BluetoothServiceDescription(service)

5.1 BatteryService

ServiceType	BluetoothSIGstandardservice
ServiceUUID	0x180F
Batterylevel data Characteristic UUID	0x2A19
Batteryleveldata permissions	ReadandNotification
Batterydatabytes	1 byte
Datacontent	0~ 100 indicatesthatthecurrentbatterylevelrangesfrom 0 to100%
Example	0x64 indicatesthatthecurrentBluetoothbatteryis 100%

Table 5

5.2 DeviceinformationService

ServiceType	BluetoothSIGstandardservice
ServiceUUID	0x180A
UUID(ManufacturerNameCharacteristic)	0x2A29
UUID(SoftwareRevisionCharacteristic)	0x2A28
UUID(HardwareRevisionCharacteristic)	0x2A27
UUID (Firmware RevisionCharacteristic)	0x2A26
UUID (Serial Number Characteristic)	0x2A25
Datareadpermission	ReadOnly
Example	ManufacturerName : UMEING

Table 6

5.3 Keyservice

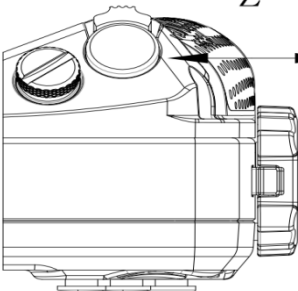
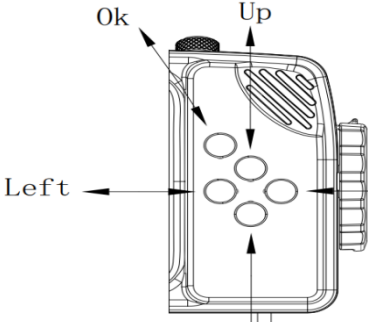
Servicetype		Bluetoothprivateservice	
ServicebasisUUID	0x23, 0xD1, 0xBC, 0xEA, 0x5F, 0x78, 0x23, 0x15, 0xDE,0xEF, 0x12, 0x12, 0xxx, 0xxx ,0x00, 0x00, 0x00, 0x00		
ServiceUUID	0x1523		
KeyCharacteristicUUID	0x1524		
Keydatapermissions	ReadandNotification		
keydatabytes	1byte		
Z - Z + Shutter  Ok Up Left Right Down 			
Keyvalue	Item	ShortPress	Longpress
	Key1(right)	0x10	
	Key2(shutter)	0x20(Power on)	
	Key3(up)	0x30	
	Key4(left)	0x40	
	Key5(ok)	0x50	Power off
	Key 6 (down)	0x61	0x60
	Key 7 (Z+)	0x70	
	Key 8 (Z-)	0x80	
Example	Bluetoothnotifiesthephonethatthereis 0x10 data, whichmeansKey1 hasbeenshort- pressed.		
Trigger flash LED Characteristic UUID	0x1525		
Trigger flash LED data permission	Write		
Trigger flash LED data byte	1byte		
	0x01(Any number)	Trigger flash once	
Example	0x01 indicates Trigger flash once		

Table7

5.4 Water pressure and temperature sensor, barometric pressure sensor data reporting, vacuuming start indication, Air extraction cover signal service

Servicetype	Bluetoothprivateservice			
ServicebasisUUID	0x23, 0xD1, 0xBC, 0xEA, 0x5F, 0x78, 0x23, 0x15,0xDE, 0xEF, 0x12, 0x12, 0xxx, 0xxx ,0x00, 0x00, 0x00,0x00			
ServiceUUID	0x1623			
Motor control characteristics UUID	0x1624			
Startdatapmission	Write			
Startdatabyte	1 byte			
	0x01	Vacuumingstart (non - 0 data)		
	0x00	VacuumingStop		
Example	0x01 indicate that vacuuming start			
Water pressure sensor characteristics UUID	0x1625			
Sensor data permission	ReadandNotification			
Sensor data bytes	4 bytes (Little-Endian)			
Sensor pressure data value: 32-bit unsigned integer (0.1mba)	1stbyte (MSB)	2ndbyte	3rdbyte	4thbyte (LSB)
	(LSB)			(MSB)
Example	0x83270000 indicates that thecurrentwater pressure is 1011.5mba (0x2783)			
Water pressure sensor temperature characteristic UUID	0x1626			
Sensor data permission	Read and Notification			
Sensor data bytes	4 bytes (Little-Endian)			
Sensor water temperature data value: 32-bit unsigned integer (0.01℃)	1stbyte (MSB)	2ndbyte	3rdbyte	4thbyte (LSB)
	(LSB)			(MSB)
Example	0x580b0000 indicates that thecurrentwatertemperature is 29.04 °C (0xb58)			
Barometric pressuresensor characteristics UUID	0x1627			
Sensor data permission	Read and Notification			
Sensor data bytes	4 bytes (Little-Endian)			
Sensor pressure data value: 32-bit unsigned integer (1 pa)	1stbyte (MSB)	2ndbyte	3rdbyte	4thbyte (LSB)
	(LSB)			(MSB)

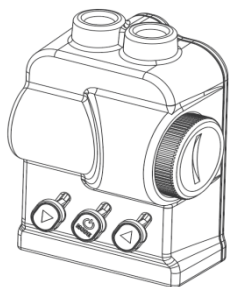
Example	0x7f8a0100 indicates that the current barometric pressure is 100991pa (0x18a7f)	
Air extraction cover signalcharacteristics UUID	0x1628	
Permissiontoreportdata	ReadandNotification	
Reporteddatabytes	1 byte	
Reported data value	0x00	Air extraction cover open
	0x01	Air extraction cover close
Example	0x01 indicates that Air extraction cover close currently	
solenoid valvecontrol characteristics UUID	0x1629	
Startdatapermission	Write	
Startdatabyte	1 byte	
	0x01	solenoid valveopen (non - 0 data)
	0x00	solenoid valve Stop
Example	0x01 indicate that solenoid valve open	
Infrared Remote Control characteristics UUID	0x162A	
Startdatapermission	Write	
Startdatabyte	1 byte	
	0x01—Increase the brightness of the flashlight (Equivalent to short pressing the left button on the remote control)	
	0x02—Switch the type of flashlight light (Equivalent to short pressing the middle button of the remote control)	
	0x03 –Reduce the brightness of the flashlight (Equivalent to short pressing the right button on the remote control)	
	0x04—Flashlight enters sleep mode (Equivalent to long pressing the middle button of the remote control)	
	0x05—The flashlight is starting to wake up (Equivalent to long pressing the middle button of the remote control again)	
	0x06—Flashlight enters focus or flashing mode (Equivalent to simultaneously pressing the left and right buttons on the remote control)	
Example	0x01The current LED brightness level of the flashlight has been increased by 1 level	

Table8